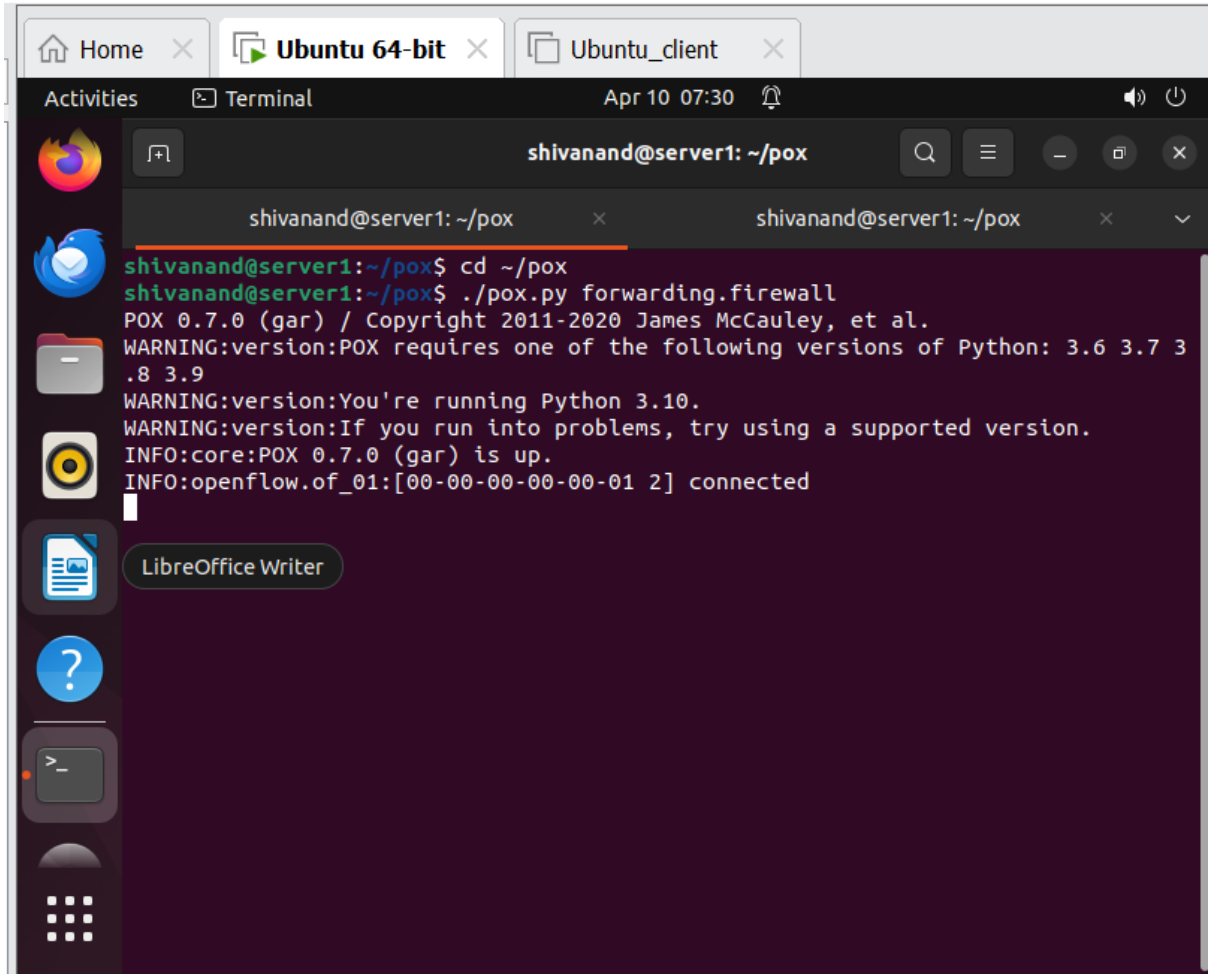
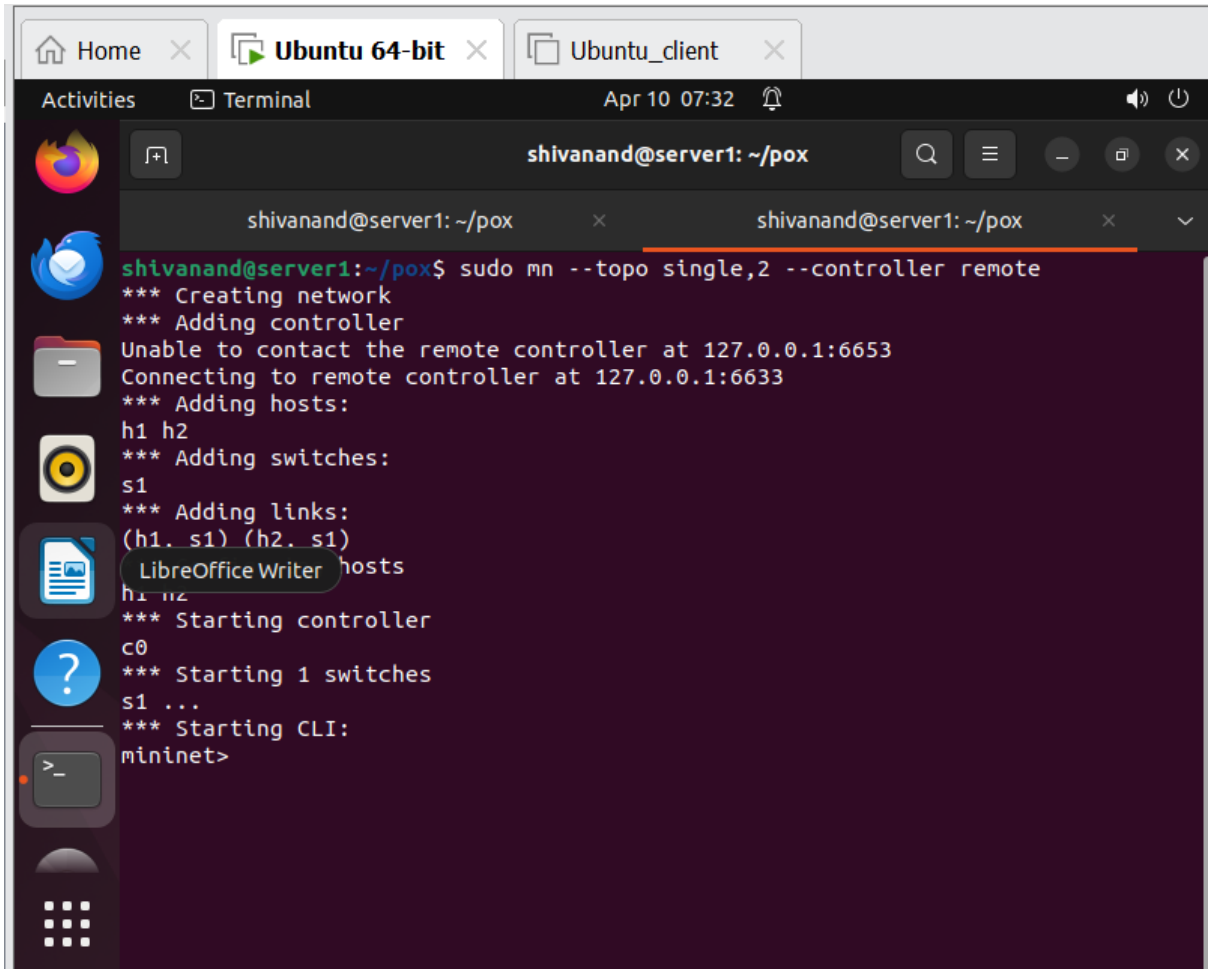
Screenshots / Proof of Execution

1. Mininet Topology

Description: Shows 2 hosts and 1 switch connected to the controller.

[Insert Screenshot Here: Mininet startup output]



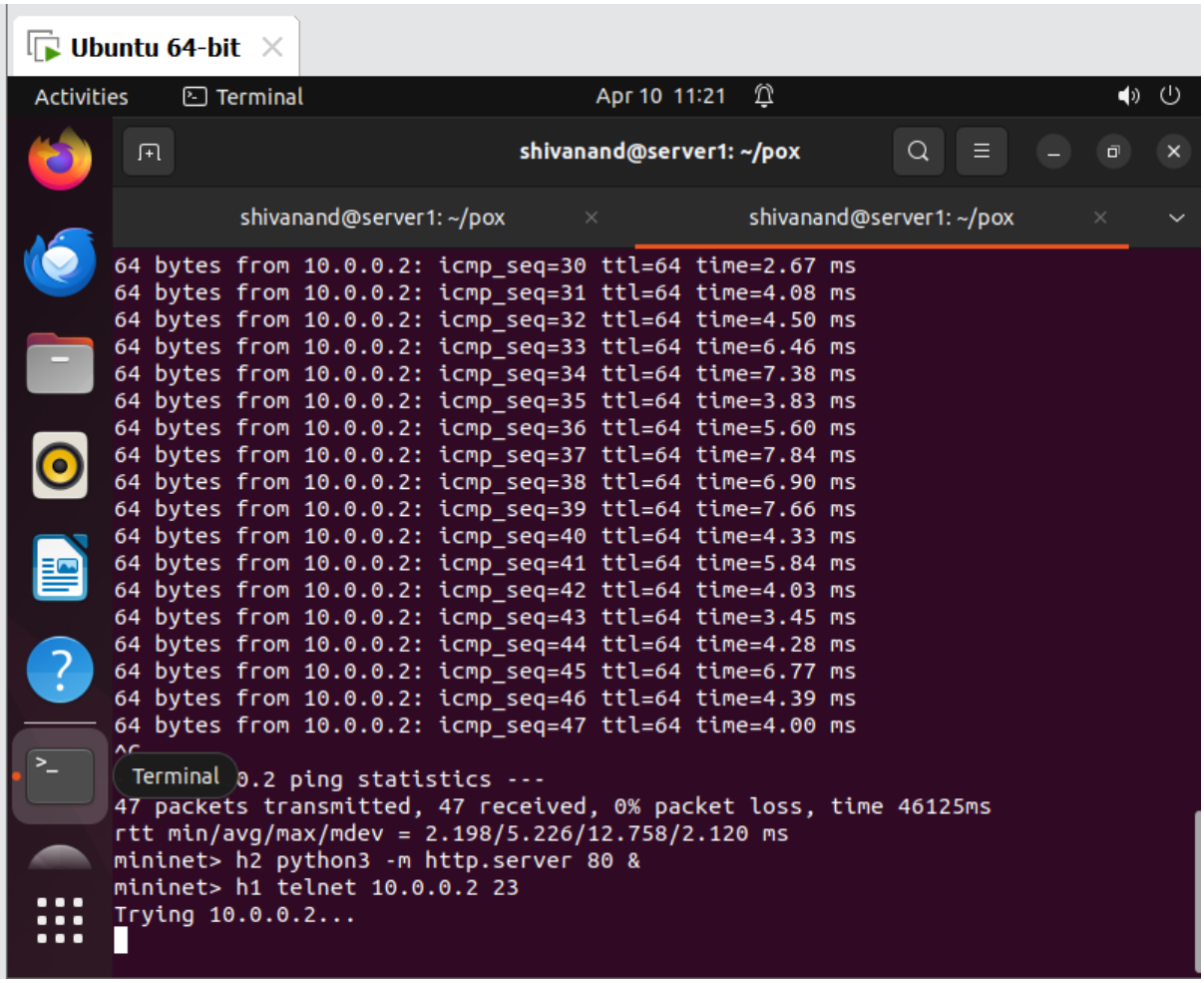


2. Ping Test

Command:
mininet> h1 ping h2

Expected Output:
Ping is successful between h1 and h2.

[Insert Screenshot Here: Ping output]

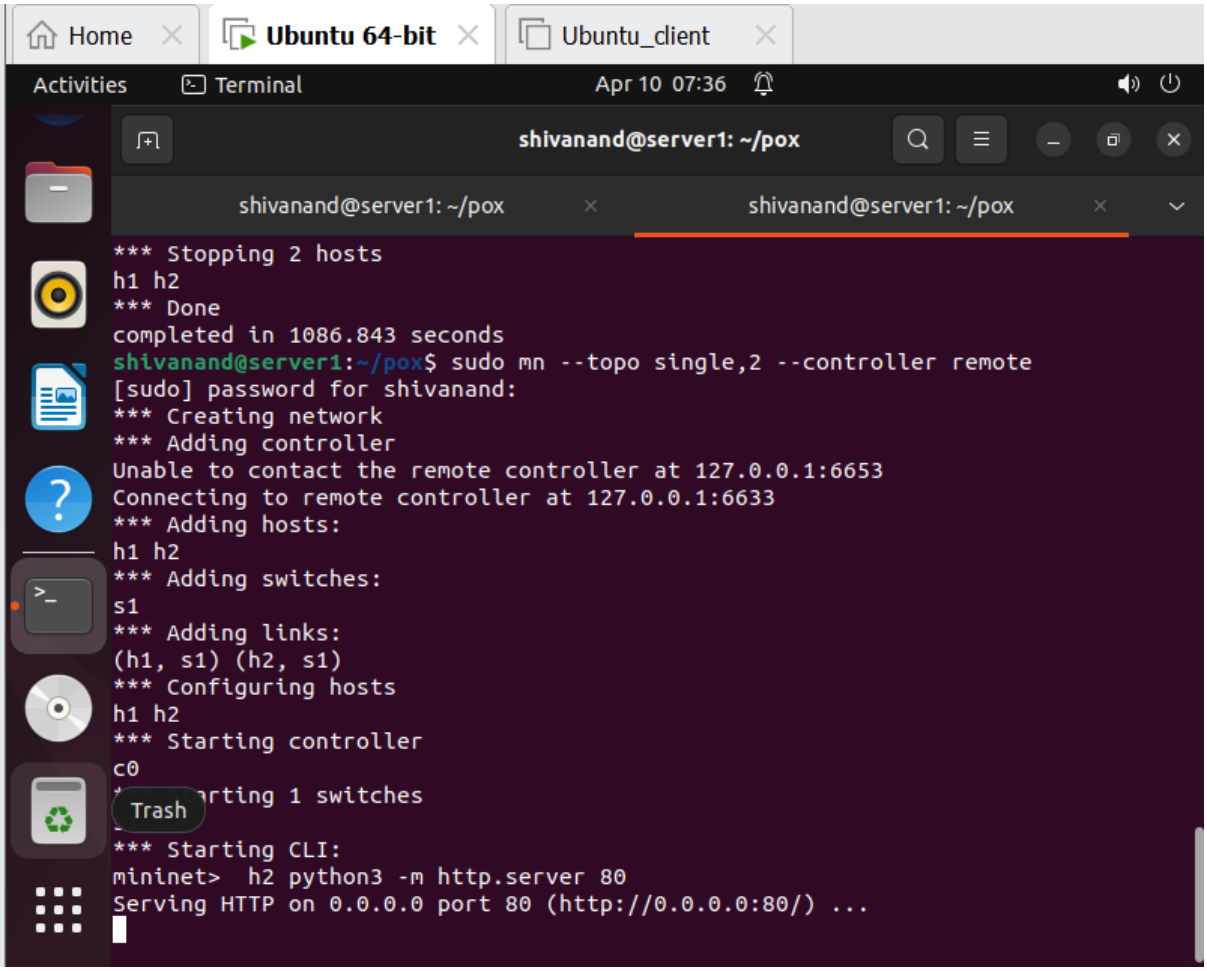


3. HTTP Test (Allowed)

Command:
mininet> h2 python3 -m http.server 80
mininet> h1 curl http://10.0.0.2

Expected Output:
HTML page is displayed, indicating HTTP traffic is allowed.

[Insert Screenshot Here: HTTP output]


Home
Ubuntu 64-bit
Ubuntu_client
Activities
Terminal
Apr 10 07:36
shivanand@server1: ~/pox
shivanand@server1: ~/pox
shivanand@server1: ~/pox
*** Stopping 2 hosts
h1 h2
*** Done
completed in 1086.843 seconds
shivanand@server1:~/pox$ sudo mn --topo single,2 --controller remote
[sudo] password for shivanand:
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
Trash
*** Starting CLI:
mininet> h2 python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...

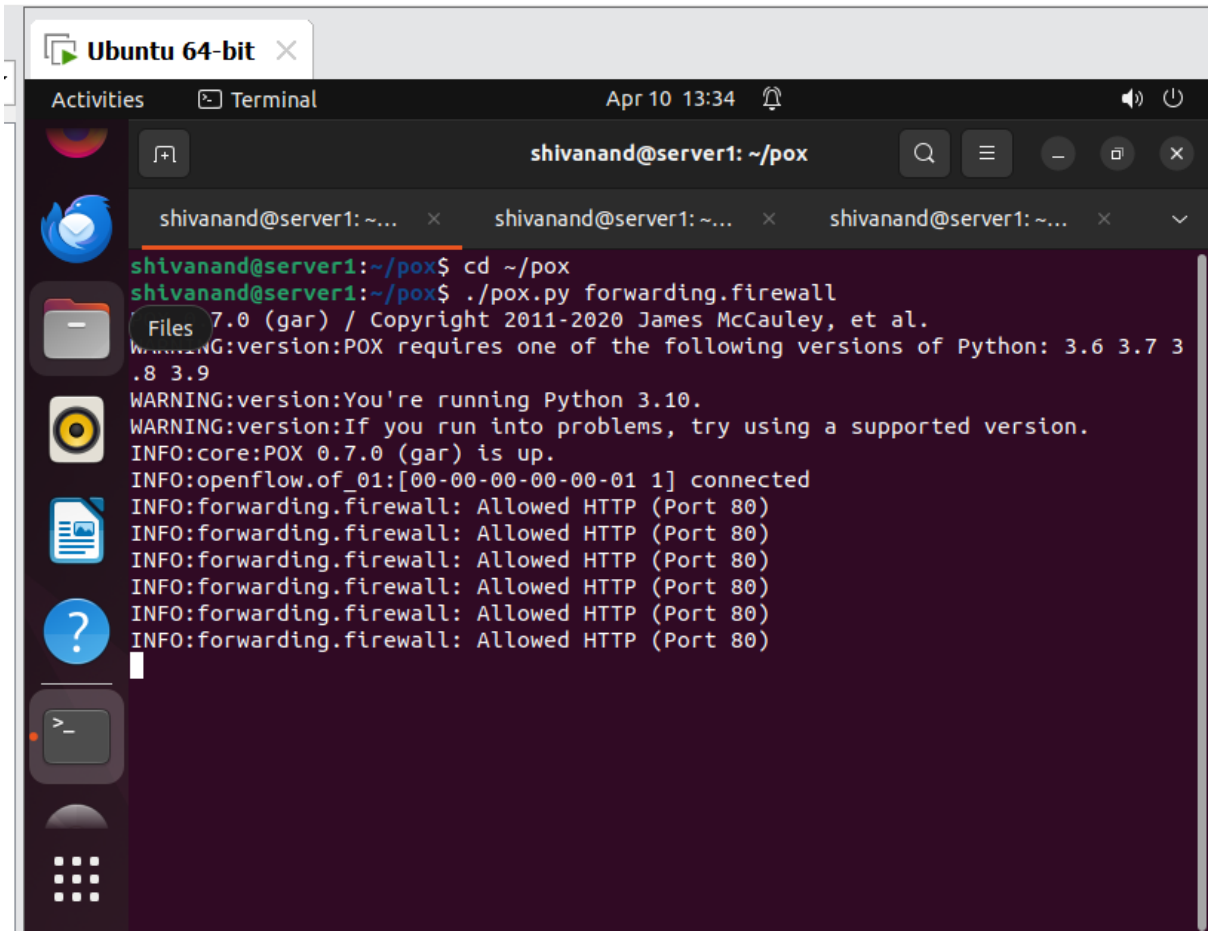
Home | Ubuntu 64-bit | Ubuntu_client

Activities | Terminal | Apr 10 07:38

shivanand@server1: ~/pox

shivanand@server1: ~/pox | shivanand@server1: ~/pox

```
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href=".git/">.git/</a></li>
<li><a href=".gitignore">.gitignore</a></li>
<li><a href=".pylint">.pylint</a></li>
<li><a href="debug-pox.py">debug-pox.py@</a></li>
<li><a href="ext/">ext/</a></li>
<li><a href="h1">h1</a></li>
<li><a href="LICENSE">LICENSE</a></li>
<li><a href="NOTICE">NOTICE</a></li>
<li><a href="pox/">pox/</a></li>
<li><a href="pox.py">pox.py</a></li>
<li><a href="README.md">README.md</a></li>
<li><a href="setup.cfg">setup.cfg</a></li>
<li><a href="tests/">tests/</a></li>
<li><a href="tools/">tools/</a></li>
</ul>
<hr>
</body>
</html>
mininet>
```

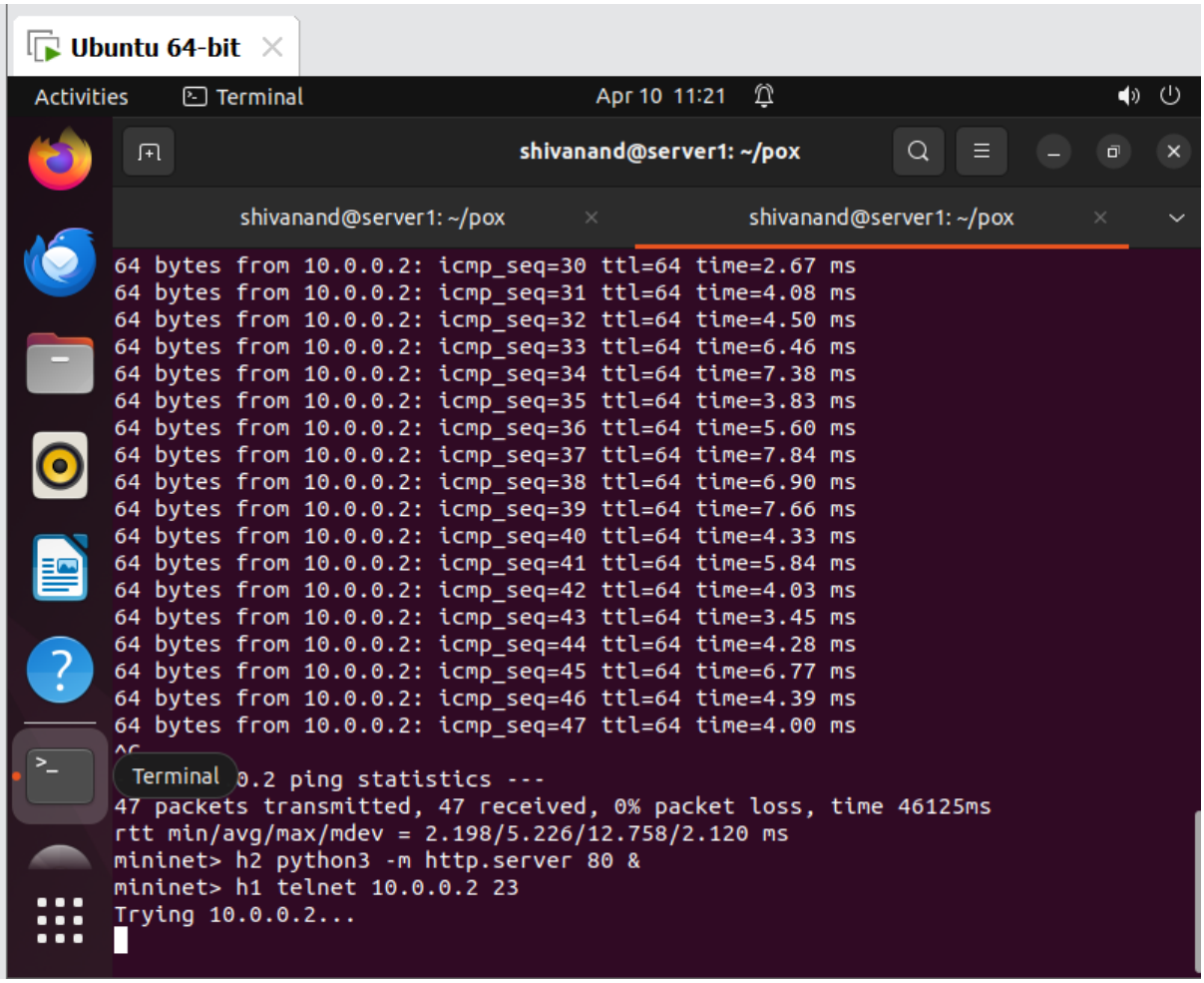


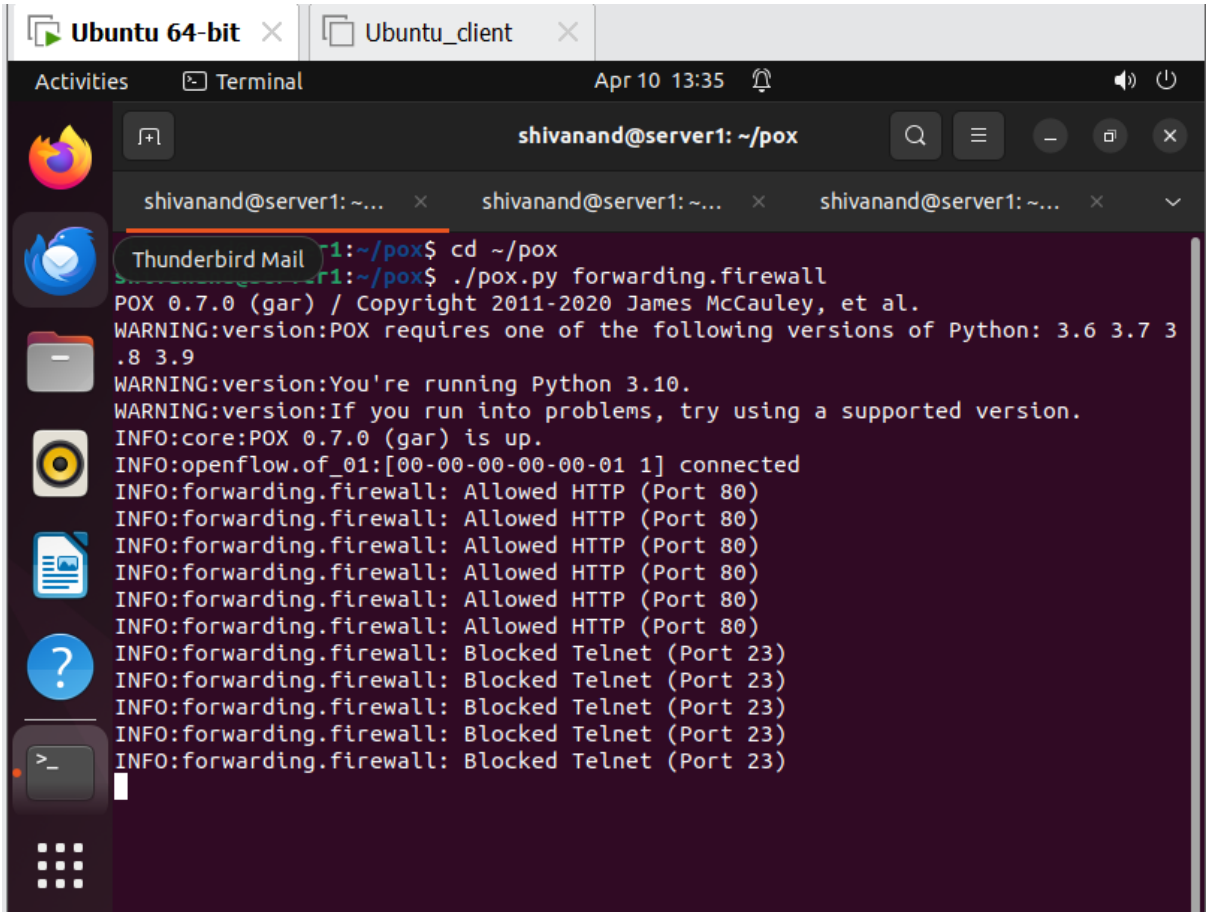
4. Telnet Test (Blocked)

Command:
mininet> h1 telnet 10.0.0.2 23

Expected Output:
Connection fails, indicating Telnet traffic is blocked.

[Insert Screenshot Here: Telnet output]


Ubuntu 64-bit
Activities
Terminal
Apr 10 11:21
shivanand@server1: ~/pox
shivanand@server1: ~/pox
shivanand@server1: ~/pox
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=2.67 ms
64 bytes from 10.0.0.2: icmp_seq=31 ttl=64 time=4.08 ms
64 bytes from 10.0.0.2: icmp_seq=32 ttl=64 time=4.50 ms
64 bytes from 10.0.0.2: icmp_seq=33 ttl=64 time=6.46 ms
64 bytes from 10.0.0.2: icmp_seq=34 ttl=64 time=7.38 ms
64 bytes from 10.0.0.2: icmp_seq=35 ttl=64 time=3.83 ms
64 bytes from 10.0.0.2: icmp_seq=36 ttl=64 time=5.60 ms
64 bytes from 10.0.0.2: icmp_seq=37 ttl=64 time=7.84 ms
64 bytes from 10.0.0.2: icmp_seq=38 ttl=64 time=6.90 ms
64 bytes from 10.0.0.2: icmp_seq=39 ttl=64 time=7.66 ms
64 bytes from 10.0.0.2: icmp_seq=40 ttl=64 time=4.33 ms
64 bytes from 10.0.0.2: icmp_seq=41 ttl=64 time=5.84 ms
64 bytes from 10.0.0.2: icmp_seq=42 ttl=64 time=4.03 ms
64 bytes from 10.0.0.2: icmp_seq=43 ttl=64 time=3.45 ms
64 bytes from 10.0.0.2: icmp_seq=44 ttl=64 time=4.28 ms
64 bytes from 10.0.0.2: icmp_seq=45 ttl=64 time=6.77 ms
64 bytes from 10.0.0.2: icmp_seq=46 ttl=64 time=4.39 ms
64 bytes from 10.0.0.2: icmp_seq=47 ttl=64 time=4.00 ms
Terminal
0.2 ping statistics ---
47 packets transmitted, 47 received, 0% packet loss, time 46125ms
rtt min/avg/max/mdev = 2.198/5.226/12.758/2.120 ms
mininet> h2 python3 -m http.server 80 &
mininet> h1 telnet 10.0.0.2 23
Trying 10.0.0.2...

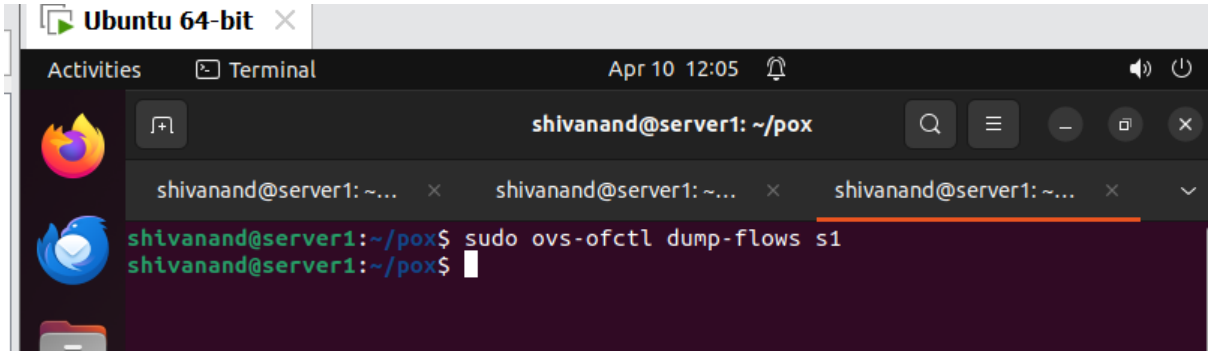


5. Flow Table Verification

Command:
sudo ovs-ofctl dump-flows s1

Expected Output:
Flow entries displayed in the switch.

[Insert Screenshot Here: Flow table]



Explanation :

The flow table shows minimal or no entries because the controller processes packets dynamically using packet_out messages instead of installing static flow rules. This behavior is expected in this implementation.”

6. Controller Logs

Description:
Shows logs for allowed and blocked traffic.

Expected Output:

- "Allowed HTTP (Port 80)"
- "Blocked Telnet (Port 23)"

[Insert Screenshot Here: POX logs]

7. Iperf Test (Optional)

Command:
mininet> iperf h1 h2

Expected Output:
Displays bandwidth between hosts.

[Insert Screenshot Here: Iperf output]

```
mininet> iperf h1 h2
*** Iperf: testing TCP bandwidth between h1 and h2
*** Results: ['25.4 Mbits/sec', '25.1 Mbits/sec']
mininet>
```